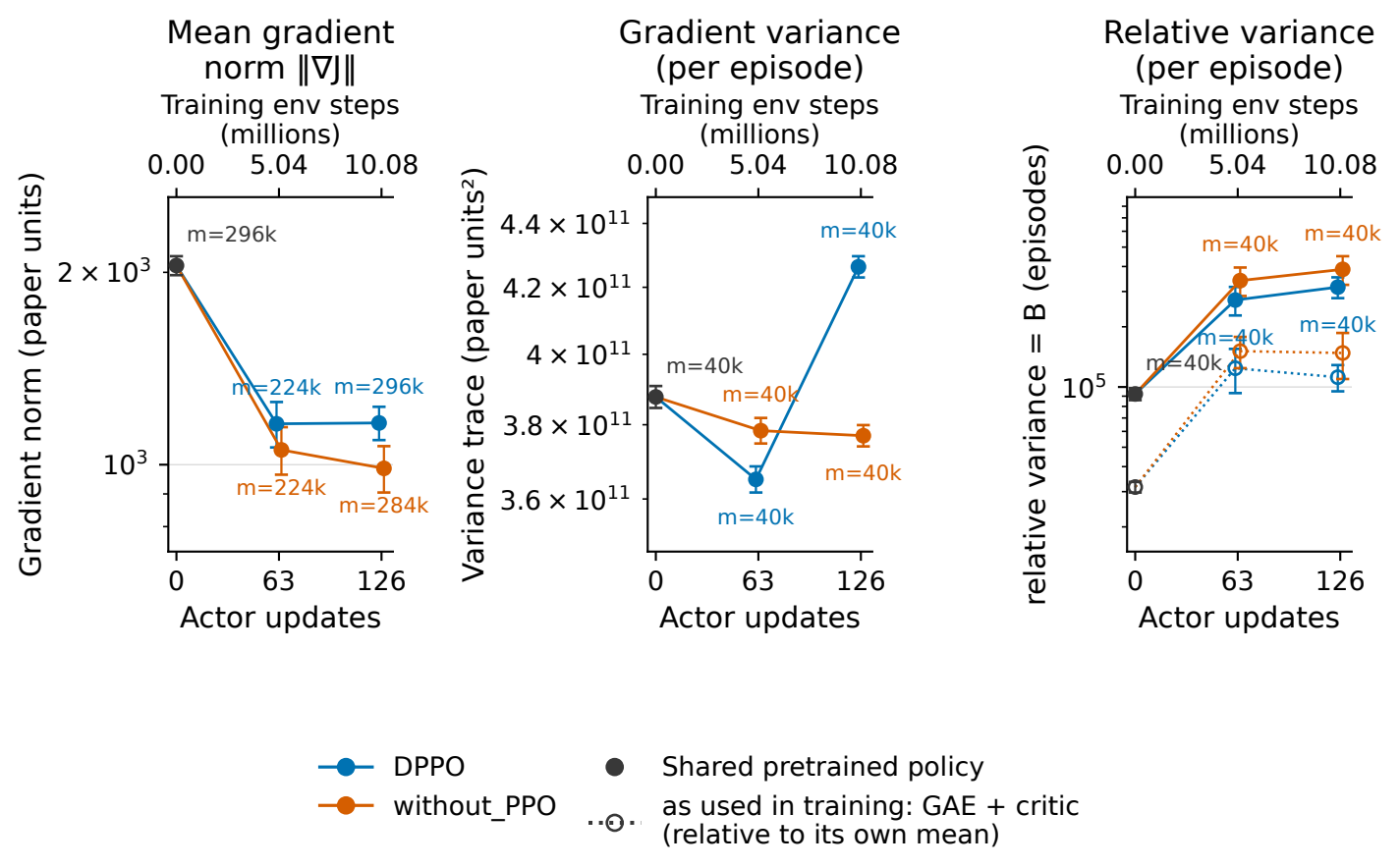
Mean gradient norm ‖∇J‖
Training env steps (millions)
0.00
5.04
10.08
Gradient norm (paper units)
2×10^3
10^3
m=296k
m=224k
m=296k
m=224k
m=284k
0
63
126
Actor updates
Gradient variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
Variance trace (paper units²)
4.4×10^{11}
4.2×10^{11}
4×10^{11}
3.8×10^{11}
3.6×10^{11}
m=40k
m=40k
m=40k
m=40k
m=40k
0
63
126
Actor updates
Relative variance (per episode)
Training env steps (millions)
0.00
5.04
10.08
relative variance = B (episodes)
10^5
m=40k
m=40k
m=40k
m=40k
m=40k
0
63
126
Actor updates
DPPO
without_PPO
Shared pretrained policy
as used in training: GAE + critic
(relative to its own mean)